

Package ‘cellchatr’

May 2, 2026

Title Fast CellChat Database Querying Backed by Rust

Version 0.1.0

Description Accelerates three bottleneck steps in the CellChat workflow using Rust via extendr. Provides parallel Wilcoxon rank-sum testing with tie correction, HashSet-based gene filtering, and ligand-receptor pair matching as drop-in replacements for CellChat functions. Achieves up to 183x speedup on Wilcoxon testing using Rayon parallelism.

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Encoding UTF-8

Roxygen list(markdown = TRUE)

RoxygenNote 7.3.3

Config/rextendr/version 0.4.2

SystemRequirements Cargo (Rust's package manager), rustc >= 1.65.0, xz

Depends R (>= 4.2)

Imports methods

Suggests knitr, rmarkdown, testthat (>= 3.0.0), CellChat, microbenchmark

Config/testthat/edition 3

VignetteBuilder knitr

NeedsCompilation yes

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hello_world	<i>Return string "Hello world!" to R.</i>
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Description

Return string "Hello world!" to R.

Usage

```
hello_world()
```

rust_match_lr_pairs	<i>Match overexpressed genes against CellChatDB ligand-receptor pairs.</i>
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Description

Match overexpressed genes against CellChatDB ligand-receptor pairs.

Usage

```
rust_match_lr_pairs(overexpressed, lr_ligands, lr_receptors, lr_names)
```

Arguments

overexpressed	Character vector of overexpressed gene names.
lr_ligands	Character vector of ligands from CellChatDB.
lr_receptors	Character vector of receptors from CellChatDB.
lr_names	Character vector of interaction names from CellChatDB.

Value

Character vector of matched interaction names.

Complexity

- Time: $O(m + r)$ where m = overexpressed genes, r = LR pairs
- Space: $O(m)$ for HashSet
- R equivalent (%%in%% twice): $O(m \times r)$

rust_subset_genes	<i>Filter expression matrix genes to only those present in CellChatDB.</i>
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Description

Filter expression matrix genes to only those present in CellChatDB.

Usage

```
rust_subset_genes(expr_genes, db_genes)
```

Arguments

expr_genes	Character vector of gene names from expression matrix.
db_genes	Character vector of gene names from CellChatDB.

Value

Character vector of genes present in both inputs.

Complexity

- Time: $O(n + m)$ where $n = \text{db_genes}$, $m = \text{expr_genes}$
- Space: $O(n)$ for HashSet
- R equivalent (`%%in%%`): $O(n \times m)$

rust_wilcoxon_filter	<i>Identify over-expressed genes per cell type using Wilcoxon rank-sum test.</i>
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Description

Identify over-expressed genes per cell type using Wilcoxon rank-sum test.

Usage

```
rust_wilcoxon_filter(counts, labels, gene_names, pval_threshold)
```

Arguments

counts	NumericMatrix of gene expression (genes x cells).
labels	Character vector of cell type labels, one per cell.
gene_names	Character vector of gene names.
pval_threshold	P-value cutoff for significance.

Value

Character vector of over-expressed gene names.

Complexity

- Time: $O((g/p) \times k \times c \log c)$ where g = genes, p = CPU cores, k = cell types, c = cells
- Space: $O(c)$ per gene
- R equivalent: $O(g \times k \times c \log c)$ — serial, no parallelism

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